

From brief to bathroom.

| A considered room, resolved through intelligence.

Turn a few human decisions into a feasible, visual bathroom plan.

Define Room dimensions, budget, design language, layout.	Resolve Catalog intelligence plus spatial constraints.
Explore Interactive 3D room and fixture inspection.	Decide Cost, space, water demand, and BOM.

CORE APPROACH

- Start with a real planning workflow.
- Use product data as the source of truth.
- Make the tradeoffs visible and explainable.
- Move from generated plan to spatial experience.

- 1. Brief
- 2. Optimize
- 3. Visualize
- 4. Inspect

Architecture that keeps decisions explainable.

User brief

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FastAPI validation

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Catalog scoring + PuLP constraints

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Wall-aware fixture placement

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+--> JSON --> Metrics + BOM

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+--> Room + fixtures --> Three.js digital twin

FastAPI boundary

CONSTRAINT AND CREATION

Typed requests, API routes, errors, static assets.

Optimizer

Theme scoring, budget, categories, clearance, placement.

Practical stack. Rich interaction.

LAYER	TECHNOLOGY	ROLE
API	FastAPI + Pydantic	Typed request boundary and JSON service
Optimization	PuLP + Pandas	Catalog scoring and constrained selection
Frontend	HTML + CSS + JavaScript	Responsive planning workflow
3D	Three.js + OBJLoader	Digital twin, assets, camera, selection
Delivery	Docker + Uvicorn	Reproducible local and container runtime

INTERACTION PRINCIPLES

- Working tool first, no marketing detour.
- Fixture HUD for focused inspection.
- Material swatches affect the active fixture.

WATER BEHAVIOR

- Confined sink and bathtub surfaces.
- Ripples, caustics, steam, vortex, splash.
- No bubbles or foam.

Constraint-aware generative merchandising.

The opportunity

Bathroom planning is fragmented across spreadsheets, product pages, floor-plan sketches, and disconnected visualizers.

The innovation

The system does not merely recommend attractive products. It resolves a collection that is budget-aware, space-aware, category-aware, placement-aware, theme-aware, and water-aware.

WHY IT MATTERS

Computational planning and human judgment meet in one loop. Users understand why a collection works, then experience it spatially through the digital twin.

Design intent in. A buildable bathroom direction out.

EXPANSION PATH

- Saved and comparable layouts
- Client-ready PDF/image export
- Accessibility and code checks
- Richer finish intelligence